

Linear Algebra

Properties of a Matrix and Operations on Matrices

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Objectives

At the end of the discussion, the learner should be able to:

- Remember and identify properties of a matrix.
- Apply properties of matrices in some logical problems.
- Perform usual binary operations on Matrices
- Perform unary operations on a matrix.



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Properties of Matrices

EQUALITY OF MATRICES

Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are said to be equal if and only if **they have the same elements**. Thus, their size or dimensions are equal as well. i.e., $a_{ij} = b_{ij}$ for all i and j .

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 5 & 2 & 3 \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 5 & 2 & 3 \end{bmatrix}$$

Therefore, $\mathbf{A} = \mathbf{B}$.



EQUALITY OF MATRICES

Example: Let $A = \begin{bmatrix} 1 & 0 \\ -4 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Given that $A = B$, find a , b , c and d .

If $A = B$, then $a = 1$, $b = 0$, $c = -4$ and $d = 2$.



EQUALITY OF MATRICES

Example: Find the values for x and y such that

$$\begin{bmatrix} x - 1 & 3 \\ 5 & -6 \\ 10 & 8 \end{bmatrix} = \begin{bmatrix} -1 & 3 \\ 5 & -6 \\ y + 4 & 8 \end{bmatrix}.$$

Solutions: Since these matrices are equal, then the following equations hold

$$x - 1 = -1 \text{ and } y + 4 = 10.$$

Hence, we obtain $x = 0$ and $y = 6$.



EQUALITY OF MATRICES

Example: Find the values for x and y if $\begin{bmatrix} 2x \\ 2x + 3y \end{bmatrix} = \begin{bmatrix} y \\ 12 \end{bmatrix}$.

Since the matrices are equal, the corresponding elements are equal!

* Form two linear equations.

$$2x = y$$

$$2x + 3y = 12$$

* Solve the system using substitution.

$$(1) \quad 2x = y$$

$$\text{From (2), } 4y = 12$$

$$\text{Substituting } y = 3 \text{ in (1), we have } 2x = 3$$

$$(2) \quad y + 3y = 12$$

$$y = 3$$

$$x = \frac{3}{2}$$



EQUALITY OF MATRICES

Exercise: Find the values for x , y and z .

$$\begin{bmatrix} x + y & z \\ y + z & x - z \end{bmatrix} = \begin{bmatrix} 7 & 3 \\ 5 & 2 \end{bmatrix}$$



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EQUALITY OF MATRICES

Like real numbers, equality of matrices also possesses the following properties:

- Matrix A is equal to itself. Hence it has **reflexive property**.
- If $A = B$, then $B = A$ for all A and B . Thus, it has a **symmetric property**.
- If $A = B$, and $B = C$, then $A = C$ for all A , B and C . Thus, it also has **transitive property**.

Therefore, matrix equality is an **equivalence relation**.



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Addition and Subtraction of Matrices

The sum or difference of two matrices, $A = [a_{ij}]$ and $B = [b_{ij}]$ of the same size yields a matrix $C = [c_{ij}]$ of the **same size** where

$$c_{ij} = a_{ij} + b_{ij} .$$

Matrices of different sizes **cannot** be added nor subtracted.

Example:

Find the sum/difference of the given pair of matrices if it exists.

$$1. A = \begin{bmatrix} -1 & 0 \\ 3 & 5 \\ -2 & 7 \end{bmatrix} + \begin{bmatrix} 4 & 4 & 1 \\ 3 & 2 & 9 \end{bmatrix}.$$

$$2. B = \begin{bmatrix} 2 & 3 & 17 & -14 & -16 \\ 5 & 10 & 24 & 13 & -12 \\ -5 & 11 & -1 & -37 & -4 \\ 0 & 19 & 18 & 20 & -1 \end{bmatrix} - \begin{bmatrix} 3 & -4 & 10 & 0 & -9 \\ -20 & 0 & -1 & 2 & -1 \\ 5 & -16 & 8 & 4 & 17 \\ -8 & 7 & 6 & 0 & 8 \end{bmatrix}.$$



Solutions:

1. Matrix A does not exist, since the summands are of different dimensions.

$$2. B = \begin{bmatrix} 2 & 3 & 17 & -14 & -16 \\ 5 & 10 & 24 & 13 & -12 \\ -5 & 11 & -1 & -37 & -4 \\ 0 & 19 & 18 & 20 & -1 \end{bmatrix} - \begin{bmatrix} 3 & -4 & 10 & 0 & -9 \\ -20 & 0 & -1 & 2 & -1 \\ 5 & -16 & 8 & 4 & 17 \\ -8 & 7 & 6 & 0 & 8 \end{bmatrix}.$$

$$= \begin{bmatrix} 2 - 3 & 3 - (-4) & 17 - 10 & -14 - 0 & -16 - (-9) \\ 5 - (-20) & 10 - 0 & 24 - (-1) & 13 - 2 & -12 - (-1) \\ -5 - 5 & 11 - (-16) & -1 - 8 & -37 - 4 & -4 - 17 \\ 0 - (-8) & 19 - 7 & 18 - 6 & 20 - 0 & -1 - 8 \end{bmatrix}.$$

continuation:

$$B = \begin{bmatrix} 2 - 3 & 3 - (-4) & 17 - 10 & -14 - 0 & -16 - (-9) \\ 5 - (-20) & 10 - 0 & 24 - (-1) & 13 - 2 & -12 - (-1) \\ -5 - 5 & 11 - (-16) & -1 - 8 & -37 - 4 & -4 - 17 \\ 0 - (-8) & 19 - 7 & 18 - 6 & 20 - 0 & -1 - 8 \end{bmatrix}.$$
$$= \begin{bmatrix} -1 & 7 & 7 & -14 & -7 \\ 25 & 10 & 25 & 11 & -11 \\ -10 & 27 & -9 & -41 & -21 \\ 8 & 12 & 12 & 20 & -9 \end{bmatrix}.$$



Example:

Find the sum/difference of the given pair of matrices if it exists.

1. $C = \begin{bmatrix} -11 & 21 & 10 \end{bmatrix} + \begin{bmatrix} 15 & -17 & 9 \end{bmatrix}$

2. $D = \begin{bmatrix} 4 & 8 \\ -6 & 5 \end{bmatrix} - \begin{bmatrix} 3 & 8 \\ 6 & 7 \end{bmatrix}$

3. $E = \begin{bmatrix} 2 & 6 & 9 \\ 5 & 4 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 6 & 9 \\ 5 & 4 & 3 \end{bmatrix}$



Solutions:

Find the sum/difference of the given pair of matrices if it exists.

$$\begin{aligned} 1. \quad C &= \begin{bmatrix} -11 & 21 & 10 \end{bmatrix} + \begin{bmatrix} 15 & -17 & 9 \end{bmatrix}. \\ &= \begin{bmatrix} -11 + 15 & 21 + (-17) & 10 + 9 \end{bmatrix}. \\ &= \begin{bmatrix} 4 & 4 & 19 \end{bmatrix}. \end{aligned}$$

$$\begin{aligned} 2. \quad D &= \begin{bmatrix} 4 & 8 \\ -6 & 5 \end{bmatrix} - \begin{bmatrix} 3 & 8 \\ 6 & 7 \end{bmatrix} \\ &= \begin{bmatrix} 4 - 3 & 8 - 8 \\ -6 - 6 & 5 - 7 \end{bmatrix}. \\ &= \begin{bmatrix} 1 & 0 \\ -12 & -2 \end{bmatrix}. \end{aligned}$$

$$\begin{aligned} 3. \quad E &= \begin{bmatrix} 2 & 6 & 9 \\ 5 & 4 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 6 & 9 \\ 5 & 4 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 2 - 2 & 6 - 6 & 9 - 9 \\ 5 - 5 & 4 - 4 & 3 - 3 \end{bmatrix}. \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \end{aligned}$$



Properties of Matrix Addition and Subtraction

Commutative Law:

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$$

$$\begin{bmatrix} 7 & 3 & -1 \\ 2 & -5 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 5 & 6 \\ -4 & -2 & 3 \end{bmatrix} = \begin{bmatrix} 8 & 8 & 5 \\ -2 & -7 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 5 & 6 \\ -4 & -2 & 3 \end{bmatrix} + \begin{bmatrix} 7 & 3 & -1 \\ 2 & -5 & 6 \end{bmatrix}.$$

Associative Law:

$$\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + \mathbf{B} + \mathbf{C}$$



Properties of Matrix Addition and Subtraction

Identity Property:

$\mathbf{A} + \mathbf{0} = \mathbf{0} + \mathbf{A} = \mathbf{A}$, where $\mathbf{0}$ denotes a zero matrix whose dimension is equal to the dimension of matrix $\mathbf{A}$.

Inverse Property for Addition:

$\mathbf{A} + (-\mathbf{A}) = \mathbf{0}$ (where $-\mathbf{A}$ is the matrix composed of $-a_{ij}$ as elements)



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Example:

Identify what property of addition is being illustrated.

$$1.) \begin{bmatrix} 3 & -4 & 1 \\ 5 & 9 & 2 \end{bmatrix} + \begin{bmatrix} 5 & 2 & 8 \\ 10 & 7 & 11 \end{bmatrix} - \begin{bmatrix} 4 & 12 & 7 \\ 7 & 4 & 9 \end{bmatrix} = \begin{bmatrix} 3 & -4 & 1 \\ 5 & 9 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 10 & 1 \\ 3 & 3 & 2 \end{bmatrix}.$$

$$2.) \begin{bmatrix} 3 & -15 \\ 11 & 7 \end{bmatrix} - \begin{bmatrix} 4 & 8 \\ 6 & 9 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 9 & 7 \end{bmatrix} = \begin{bmatrix} 3 & -15 \\ 11 & 7 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 9 & 7 \end{bmatrix} - \begin{bmatrix} 4 & 8 \\ 6 & 9 \end{bmatrix}.$$



Answer:

Identify what property of addition is being illustrated.

1.)
$$\begin{bmatrix} 3 & -4 & 1 \\ 5 & 9 & 2 \end{bmatrix} + \begin{bmatrix} 5 & 2 & 8 \\ 10 & 7 & 11 \end{bmatrix} - \begin{bmatrix} 4 & 12 & 7 \\ 7 & 4 & 9 \end{bmatrix} = \begin{bmatrix} 3 & -4 & 1 \\ 5 & 9 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 10 & 1 \\ 3 & 3 & 2 \end{bmatrix}.$$

Associative property

2.)
$$\begin{bmatrix} 3 & -15 \\ 11 & 7 \end{bmatrix} - \begin{bmatrix} 4 & 8 \\ 6 & 9 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 9 & 7 \end{bmatrix} = \begin{bmatrix} 3 & -15 \\ 11 & 7 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 9 & 7 \end{bmatrix} - \begin{bmatrix} 4 & 8 \\ 6 & 9 \end{bmatrix}.$$

Commutative property



Scalar Multiplication of a Matrix

Matrices can be multiplied by a scalar (constant or single element).

Let k be a scalar quantity and $A = [a_{ij}]$; then
$$kA = [ka_{ij}].$$

Example. If $k=4$ and $A = \begin{bmatrix} 3 & -1 \\ 2 & 1 \\ 2 & -3 \\ 4 & 1 \end{bmatrix}$, what is $4A$?



Scalar Multiplication of a Matrix

Example. If $k=4$ and $A = \begin{bmatrix} 3 & -1 \\ 2 & 1 \\ 2 & -3 \\ 4 & 1 \end{bmatrix}$ then $4 \times \begin{bmatrix} 3 & -1 \\ 2 & 1 \\ 2 & -3 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ 2 & 1 \\ 2 & -3 \\ 4 & 1 \end{bmatrix} \times 4 = \begin{bmatrix} 12 & -4 \\ 8 & 4 \\ 8 & -12 \\ 16 & 4 \end{bmatrix}$



Example

Find the product of the following:

$$1. \quad 4 \cdot \begin{bmatrix} 3 & 5 \\ -1 & 7 \\ 0 & 6 \\ 5 & -2 \end{bmatrix}$$

$$2. \quad -7 \cdot \begin{bmatrix} 6 & 9 & 7 \\ -2 & 8 & 3 \\ 1 & 0 & -1 \end{bmatrix}$$

$$3. \quad 5 \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Solutions:

Find the product of the following:

$$1. \quad 4 \cdot \begin{bmatrix} 3 & 5 \\ -1 & 7 \\ 0 & 6 \\ 5 & -2 \end{bmatrix} = \begin{bmatrix} 12 & 20 \\ -4 & 28 \\ 0 & 24 \\ 20 & -8 \end{bmatrix}.$$

$$2. \quad -7 \cdot \begin{bmatrix} 6 & 9 & 7 \\ -2 & 8 & 3 \\ 1 & 0 & -1 \end{bmatrix} = \begin{bmatrix} -42 & -63 & -49 \\ 14 & -56 & -21 \\ -7 & 0 & 7 \end{bmatrix}.$$

$$3. \quad 5 \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

Remark: Every scalar matrix is a product of a scalar number and an identity matrix.



Properties of Scalar Multiplication of a Matrix

Let k and g be any constant numbers and suppose A and B are two matrices of the same dimensions, then

- $k(A + B) = kA + kB$. (Distributive Property)
- $(k + g)A = kA + gA$. (Distributive Property)
- $k(AB) = (kA)B = A(kB)$. (Associative Property)
- $k(gA) = (kg)A$. (Associative Property)



MULTIPLICATION OF MATRICES

Consider two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$. The product $A \cdot B$ exists if the number of columns of A must equal the number of rows of B .

Example: Suppose matrix $A = [a_{ij}]$ dimensions 3×4 , then matrix $B = [b_{ij}]$ must have a dimensions $4 \times n$ for any positive integer n so that $A \cdot B$ exists.

Thus, in general if matrix $A = [a_{ij}]$ has dimensions $m \times p$, then matrix $B = [b_{ij}]$ must have dimensions $p \times n$ for any positive integers m, n , and p so that matrix multiplication can be perform.



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Example

From the given matrices below, find pairs of distinct matrices such that their product exists. Express your answers in matrix multiplication form.

$$A = \begin{bmatrix} 5 & 9 & 1 \end{bmatrix} \quad B = \begin{bmatrix} -2 & 1 \\ 3 & 0 \\ 8 & 2 \end{bmatrix} \quad C = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \quad D = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}$$

$$E = \begin{bmatrix} 0 & -1 & 5 & 1 & 0 \\ 4 & 2 & 3 & -4 & 4 \\ 1 & 3 & -1 & 1 & 8 \end{bmatrix} \quad F = \begin{bmatrix} 2 & 6 \\ -3 & 1 \\ 4 & 7 \\ 5 & 0 \end{bmatrix}$$



Answers:

- $A \cdot B$
- $A \cdot E$
- $B \cdot C$
- $B \cdot D$
- $C \cdot A$
- $D \cdot C$
- $E \cdot F$
- $F \cdot C$
- $F \cdot D$



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Size of the Product

If matrix $A = [a_{ij}]$ has dimensions $m \times p$ and matrix $B = [b_{ij}]$ has dimensions $p \times n$ for any positive integers m, n , and p , then product $A \cdot B$ exists. Moreover, the dimensions of $A \cdot B$ must be $m \times n$ where m is the number of rows in A and n is the number of columns in B .



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Example

From the previous example, we these given matrices:

$$A = \begin{bmatrix} 5 & 9 & 1 \end{bmatrix} \quad B = \begin{bmatrix} -2 & 1 \\ 3 & 0 \\ 8 & 2 \end{bmatrix} \quad C = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \quad D = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}$$

$$E = \begin{bmatrix} 0 & -1 & 5 & 1 & 0 \\ 4 & 2 & 3 & -4 & 4 \\ 1 & 3 & -1 & 1 & 8 \end{bmatrix} \quad F = \begin{bmatrix} 2 & 6 \\ -3 & 1 \\ 4 & 7 \\ 5 & 0 \end{bmatrix}$$

Also, we have verified that the following products exist:

- $A \cdot B$
- $A \cdot E$
- $B \cdot C$
- $B \cdot D$
- $C \cdot A$
- $D \cdot C$
- $E \cdot F$
- $F \cdot C$
- $F \cdot D$

Find the dimensions for each product

Answers:

Product	Dimensions
• $A \cdot B$	1×2
• $A \cdot E$	1×4
• $B \cdot C$	3×1
• $B \cdot D$	3×2
• $C \cdot A$	2×3
• $D \cdot C$	2×1
• $E \cdot F$	3×2
• $F \cdot C$	4×1
• $F \cdot D$	4×2

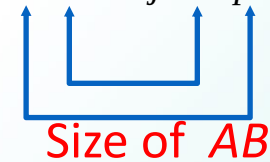


MULTIPLICATION OF MATRICES

The product of two matrices exists if and only if the given matrices are conformable for multiplication.

If $A = [a_{ij}]_{m \times n}$, $B = [b_{ij}]_{n \times p}$. Then $AB = [a_{ij}]_{m \times n} [b_{ij}]_{n \times p} = [c_{ij}]_{m \times p}$

where



$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} b_{11} & \cdots & b_{1j} & \cdots & b_{1n} \\ b_{21} & \vdots & b_{2j} & \cdots & b_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ b_{n1} & \cdots & b_{nj} & \cdots & b_{nn} \end{bmatrix} = \begin{bmatrix} c_{i1} & c_{i2} & \cdots & c_{ij} & \cdots & c_{in} \end{bmatrix}$$

Operations on Matrices

Example1:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}$$

where

$$(a_{11} \times b_{11}) + (a_{12} \times b_{21}) + (a_{13} \times b_{31}) = c_{11}$$

$$(a_{11} \times b_{12}) + (a_{12} \times b_{22}) + (a_{13} \times b_{32}) = c_{12}$$

$$(a_{21} \times b_{11}) + (a_{22} \times b_{21}) + (a_{23} \times b_{31}) = c_{21}$$

$$(a_{21} \times b_{12}) + (a_{22} \times b_{22}) + (a_{23} \times b_{32}) = c_{22}$$

Successive multiplication of row i of **A** with column j of **B** – row by column multiplication



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Operations on Matrices

Example 2:
$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 2 & 7 \end{bmatrix} \begin{bmatrix} 4 & 8 \\ 6 & 2 \\ 5 & 3 \end{bmatrix} = \begin{bmatrix} (1 \times 4) + (2 \times 6) + (3 \times 5) & (1 \times 8) + (2 \times 2) + (3 \times 3) \\ (4 \times 4) + (2 \times 6) + (7 \times 5) & (4 \times 8) + (2 \times 2) + (7 \times 3) \end{bmatrix}$$
$$= \begin{bmatrix} 31 & 21 \\ 63 & 57 \end{bmatrix}$$

Example 3:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 31 & 21 \\ 63 & 57 \end{bmatrix} = \begin{bmatrix} 31 & 21 \\ 63 & 57 \end{bmatrix}$$



Example

$$\begin{aligned} [5 \quad 9 \quad 1] \cdot \begin{bmatrix} -2 & 1 \\ 3 & 0 \\ 8 & 2 \end{bmatrix} &= [5(-2) + 9(3) + 1(8) \quad 5(1) + 9(0) + 1(2)]. \\ &= [-10 + 27 + 8 \quad 5 + 0 + 2]. \\ &= [25 \quad 7]. \end{aligned}$$

Example

$$\begin{bmatrix} 2 & 6 \\ -3 & 1 \\ 4 & 7 \\ 5 & 0 \end{bmatrix} \cdot \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2(2) + 6(1) & 2(-1) + 6(2) \\ -3(2) + 1(1) & -3(-1) + 1(2) \\ 4(2) + 7(1) & 4(-1) + 7(2) \\ 5(2) + 0(1) & 5(-1) + 0(2) \end{bmatrix}.$$
$$= \begin{bmatrix} 10 & 10 \\ -5 & 5 \\ 15 & 10 \\ 10 & -5 \end{bmatrix}.$$



Properties for Multiplication of Matrices

Assuming that matrices A, B and C are conformable for the operations indicated, then the following are true:

1. $AI = IA = A$
2. $A(BC) = (AB)C = ABC$ - **(associative law)**
3. $A(B+C) = AB + AC$ - **(first distributive law)**
4. $(A+B)C = AC + BC$ - **(second distributive law)**

Remarks:

1. AB not generally equal to BA, BA may not be conformable
2. If $AB = 0$, neither A nor B necessarily = 0
3. If $AB = AC$, B not necessarily = C



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Example

Perform the indicated operations to simplify the given matrices.

$$2 \begin{bmatrix} 5 & 1 \\ 6 & 3 \end{bmatrix} - 3 \begin{bmatrix} 1 & 2 \\ 0 & 4 \end{bmatrix} + \begin{bmatrix} 3 & -1 & 4 & 7 \\ 6 & 1 & 5 & 2 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 \\ 8 & 9 \\ -3 & 1 \\ 1 & -3 \end{bmatrix}$$



Solution:

$$2 \begin{bmatrix} 5 & 1 \\ 6 & 3 \end{bmatrix} - 3 \begin{bmatrix} 1 & 2 \\ 0 & 4 \end{bmatrix} + \begin{bmatrix} 3 & -1 & 4 & 7 \\ 6 & 1 & 5 & 2 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 \\ 8 & 9 \\ -3 & 1 \\ 1 & -3 \end{bmatrix}$$

Let us simplify the given by performing scalar multiplication and matrix multiplication.
We obtain

$$\left(\begin{bmatrix} 10 & 2 \\ 12 & 6 \end{bmatrix} - \begin{bmatrix} 3 & 6 \\ 0 & 12 \end{bmatrix} \right) + \begin{bmatrix} 3(2) + (-1)(8) + 4(-3) + 7(1) & 3(0) + (-1)(9) + 4(1) + 7(-3) \\ 6(2) + (1)(8) + 5(-3) + 2(1) & 6(0) + (1)(9) + 5(1) + 2(-3) \end{bmatrix}$$

Next, we find the difference of the grouped matrices and simplify the obtained product

$$\begin{bmatrix} 7 & -4 \\ 12 & -6 \end{bmatrix} + \begin{bmatrix} -7 & -26 \\ 7 & 8 \end{bmatrix}$$

Finally, we have

$$\begin{bmatrix} 0 & -30 \\ 19 & 2 \end{bmatrix}$$

Trace of a Matrix

Let $A = [a_{ij}]$ be a square matrix of order n . Then the **trace of matrix A** , denoted by $\text{tr}A$ is the sum of all diagonal entries of A . That is,

$$\text{tr}A = \sum_{i=1}^n a_{ii}.$$



Example

Find the trace of the following matrices.

1. $A = \begin{bmatrix} 5 & 1 \\ 6 & 3 \end{bmatrix}$

2. $B = \begin{bmatrix} 2 & -9 & 11 \\ -3 & 11 & 6 \\ -8 & 5 & -7 \end{bmatrix}$

3. $C = \begin{bmatrix} -4 & 3 & 0 & 6 \\ 4 & 5 & 1 & 3 \\ 1 & 6 & 9 & 5 \\ 0 & 2 & 2 & -1 \end{bmatrix}$



Solutions

$$\begin{aligned} 1. \operatorname{tr} A &= 5 + 3. \\ &= 8. \end{aligned}$$

$$\begin{aligned} 2. \operatorname{tr} B &= 2 + 11 + (-7). \\ &= 6. \end{aligned}$$

$$\begin{aligned} 3. \operatorname{tr} C &= -4 + 5 + 9 + (-1). \\ &= 9. \end{aligned}$$



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